VEPHLA UNI

TASK 10B

BY

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- Using **ANY.RUN**, explain the step-by-step process for analyzing a malware sample, including the identification of malicious indicators, behavioral analysis, and network activity inspection. Additionally, outline the methodology for developing a comprehensive technical report by analyzing a previous analyzed file which is said to be malicious and another with no threats detected

Navigating and Using ANY.RUN: A Step-by-Step Walkthrough

To begin using ANY.RUN, the user must first open their browser and search for "any run" on Google. This will lead them to the official homepage of ANY.RUN. On the homepage, they will

see a top navigation bar. Under the “Product” tab, they must locate and click on **Intelligent Threat Feed**.

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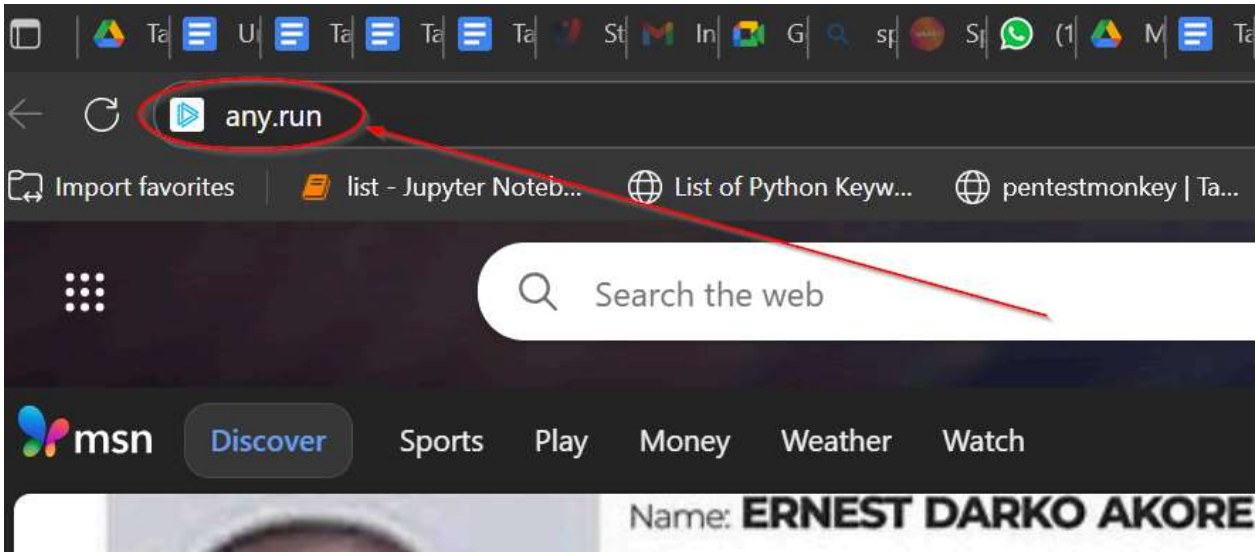
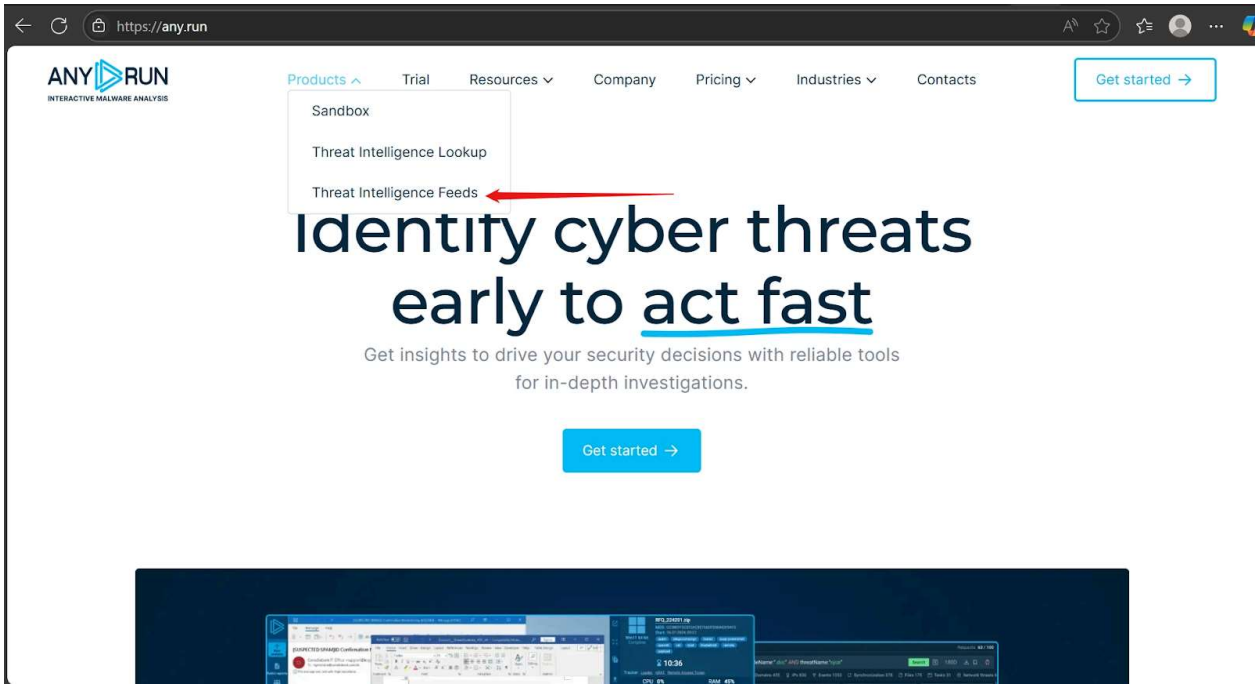
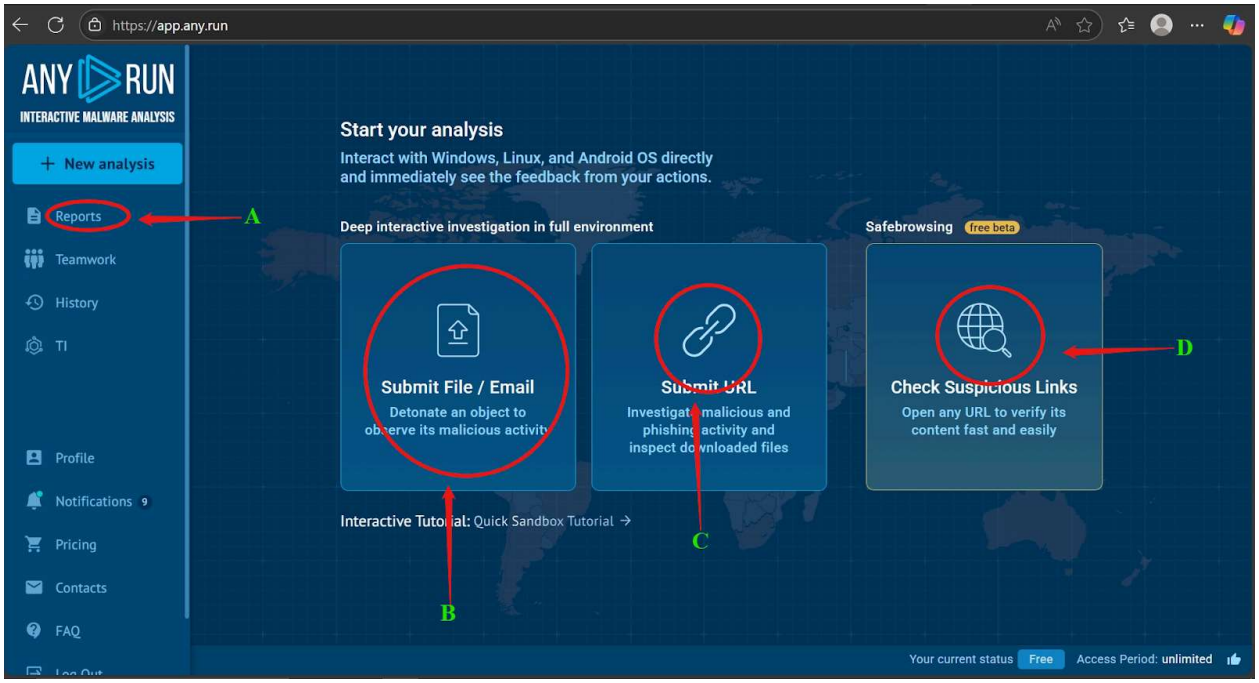




Upon clicking this option, a **Sign In** page will appear. ANY.RUN requires users to sign in with either a **business email** or a **school email**. Once signed in, the user is directed to the main dashboard or homepage of the platform.

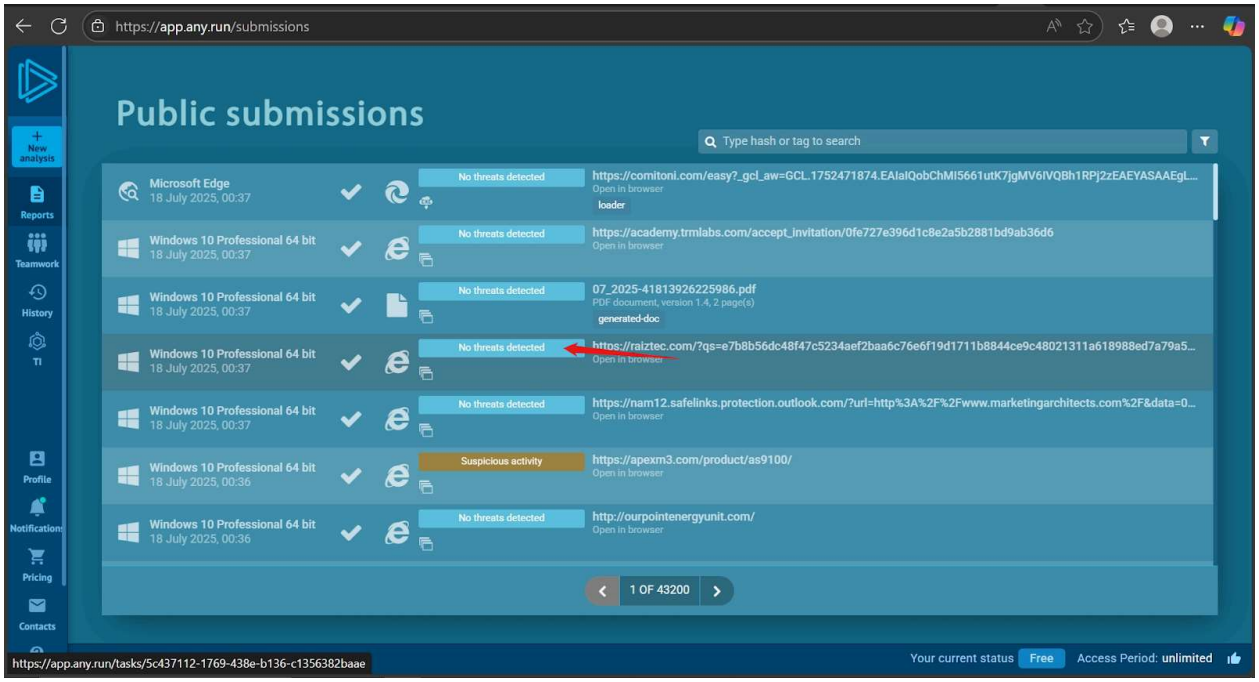


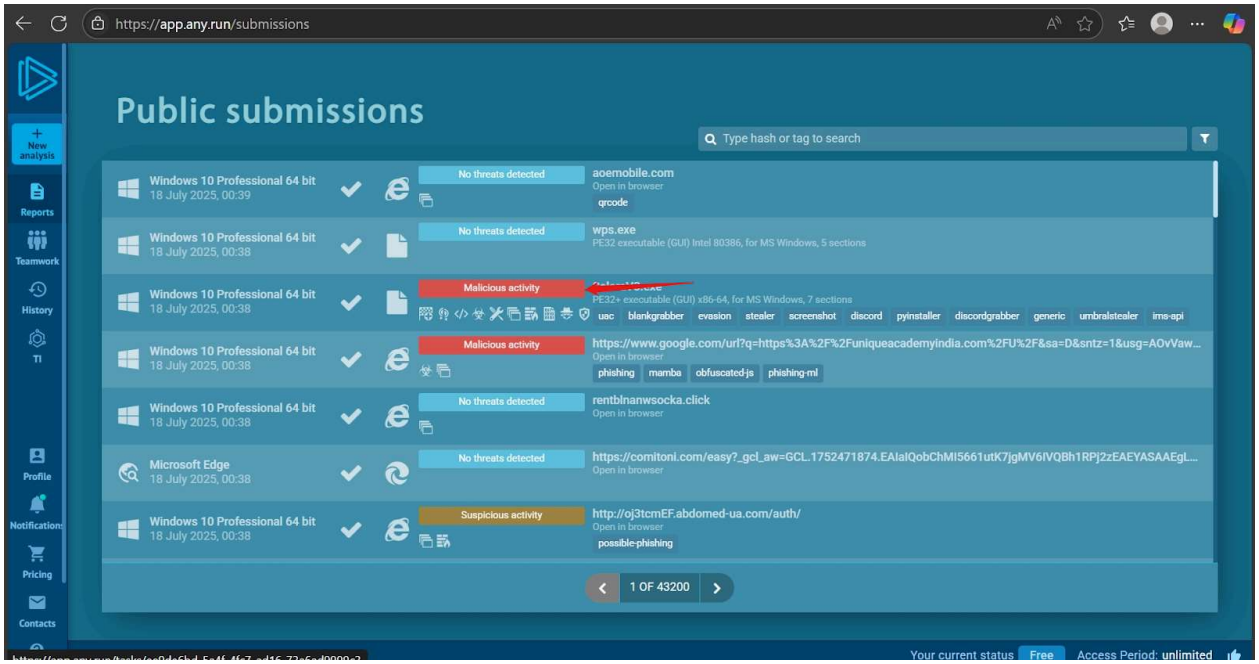
On this dashboard, users can perform a variety of functions. Towards the **left-hand side**, there is a **Reports** section where previously analyzed malware samples are listed. The dashboard also includes tools where users can **upload files** or paste **suspicious URLs** for analysis. These inputs are used to check whether a file or link is **malicious**.

Once a file or URL is submitted, users can explore previously analyzed reports. There are generally **two classifications** of outcomes:

1. **Malicious Samples –** These are highlighted in **red** to indicate a threat.

2. **Non-Threat Samples –** These are labeled as safe or clean.

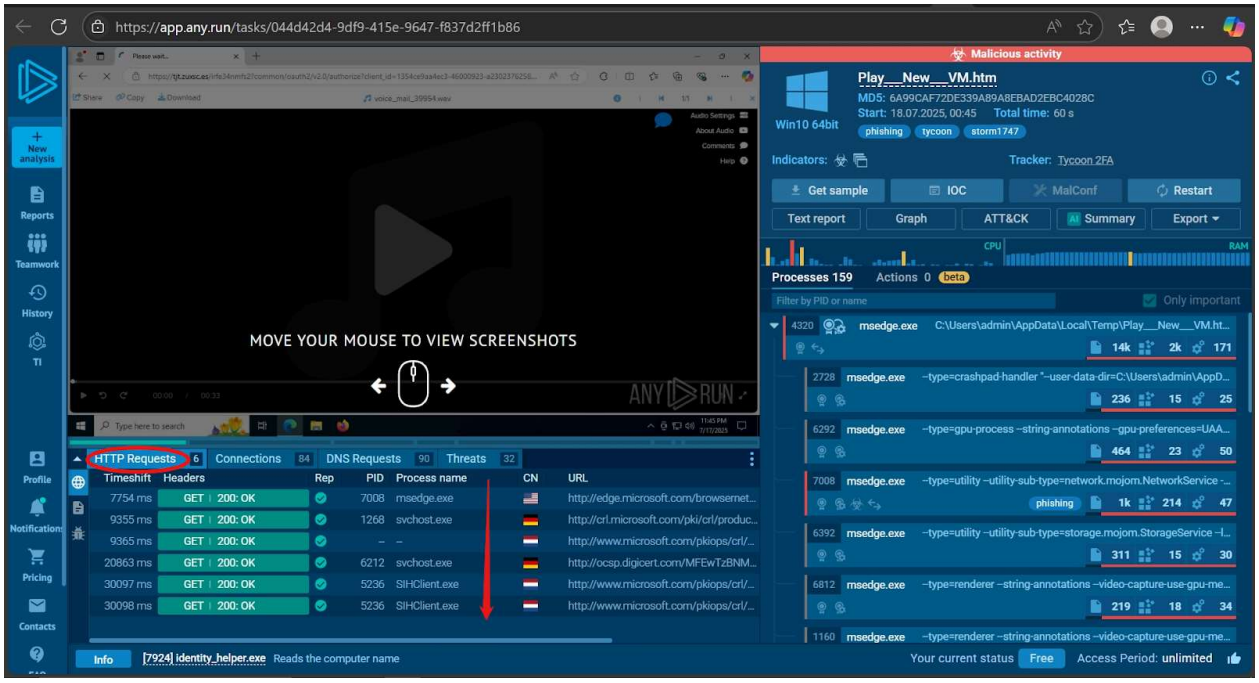


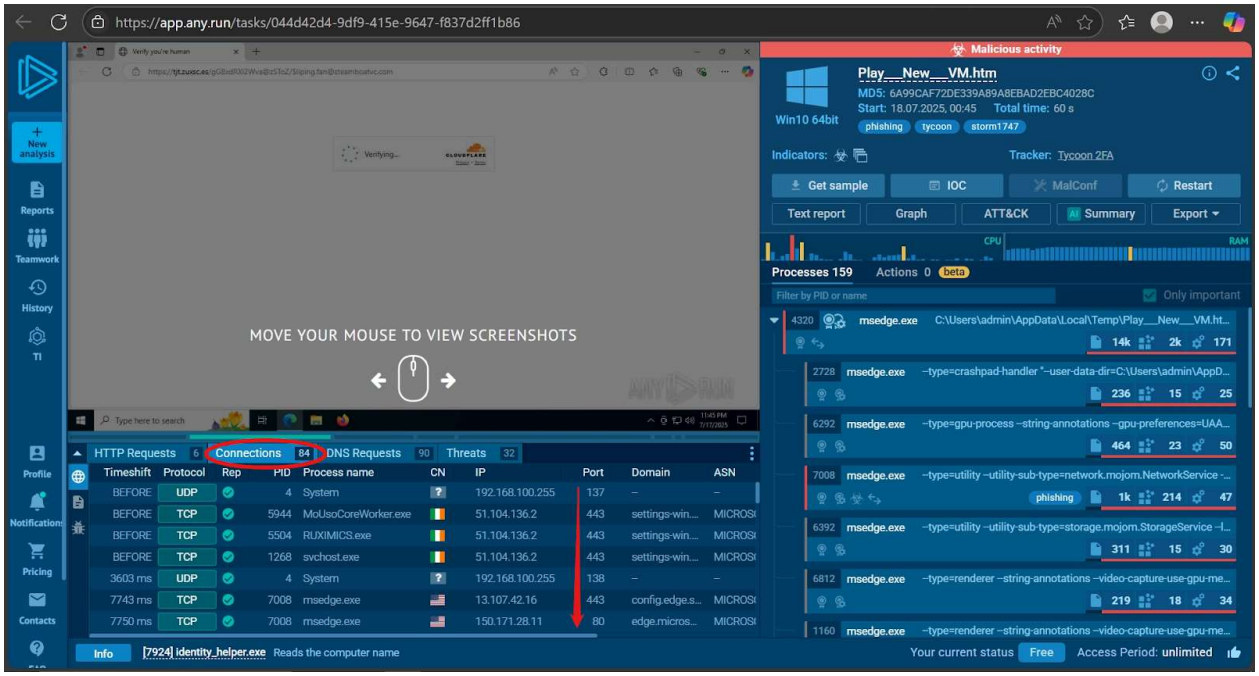


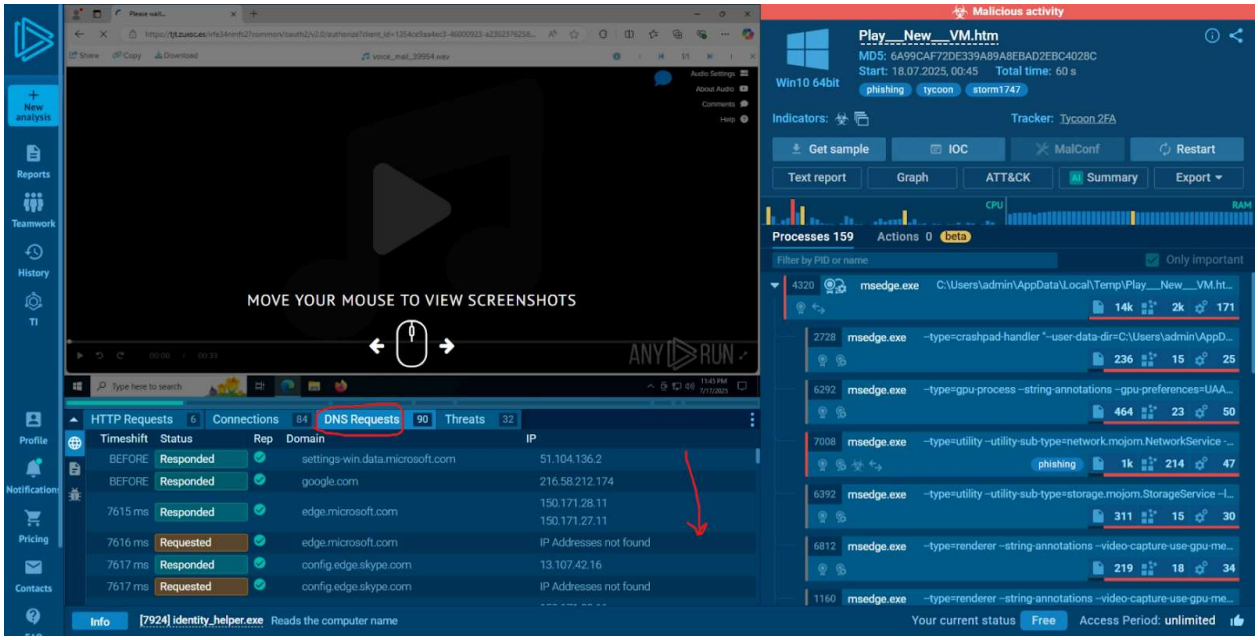
Clicking on a **malicious sample** will open a detailed report page. On this page, the user can observe various technical insights such as:

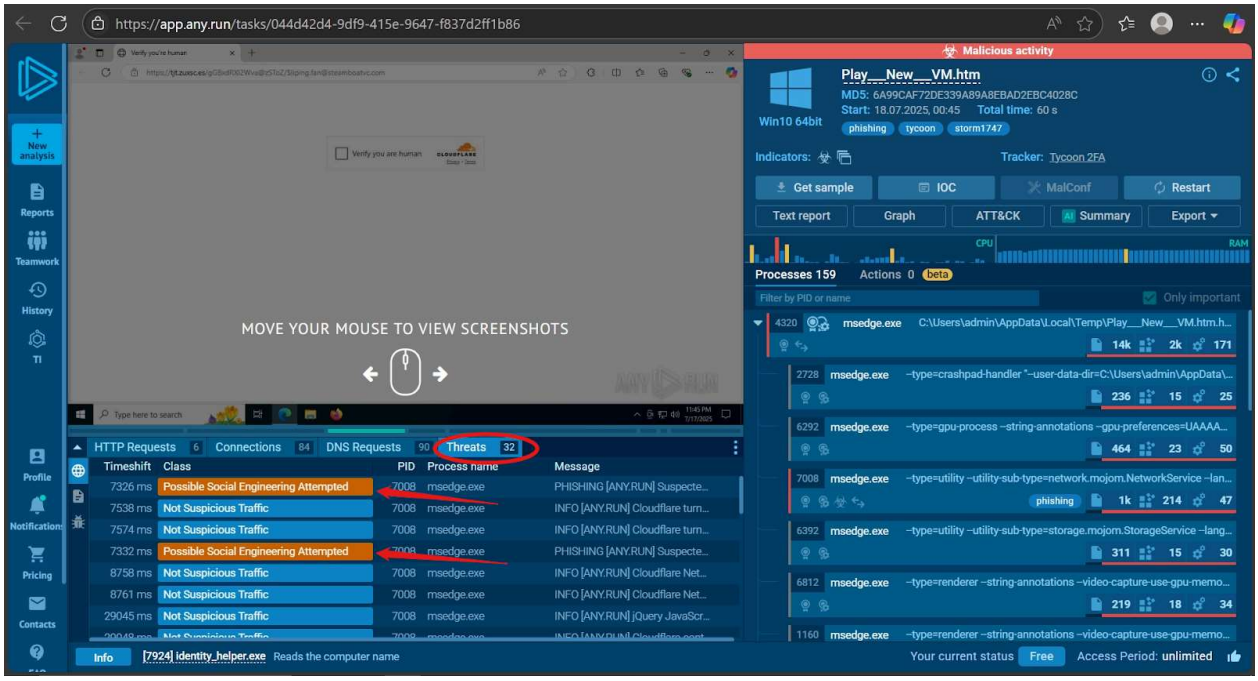
- **HTTPS connections made by the malware**
- **DNS servers contacted**

- **IP addresses accessed**
- **Request patterns**





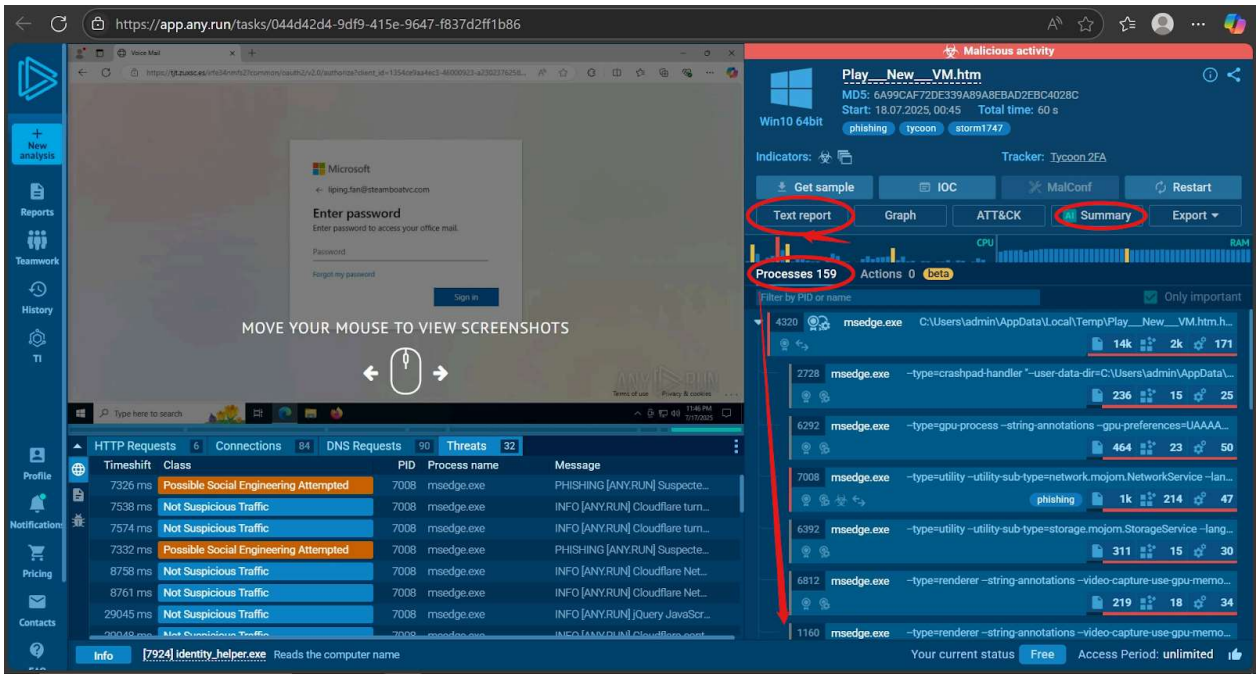




The analysis also provides threat identification. For instance, a report might classify the sample as a **social engineering attack**, possibly originating from phishing emails or similar tactics.

Users can choose to **download** the full report. The report contains:

- **General behavior of the malware**
- **Technical metadata**, such as file name, upload and analysis date
- **Indicators of compromise** like registry changes or file system manipulation
- **Execution paths and process trees**
- Whether it was flagged as **malicious** or **non-malicious**



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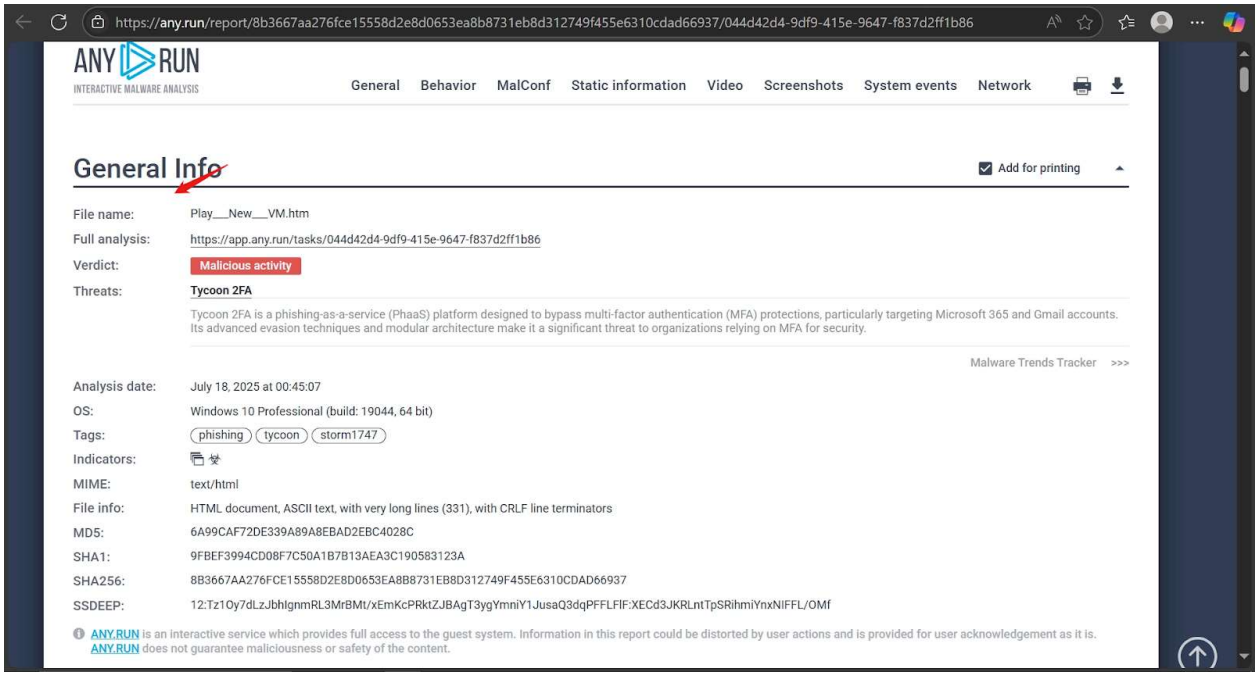
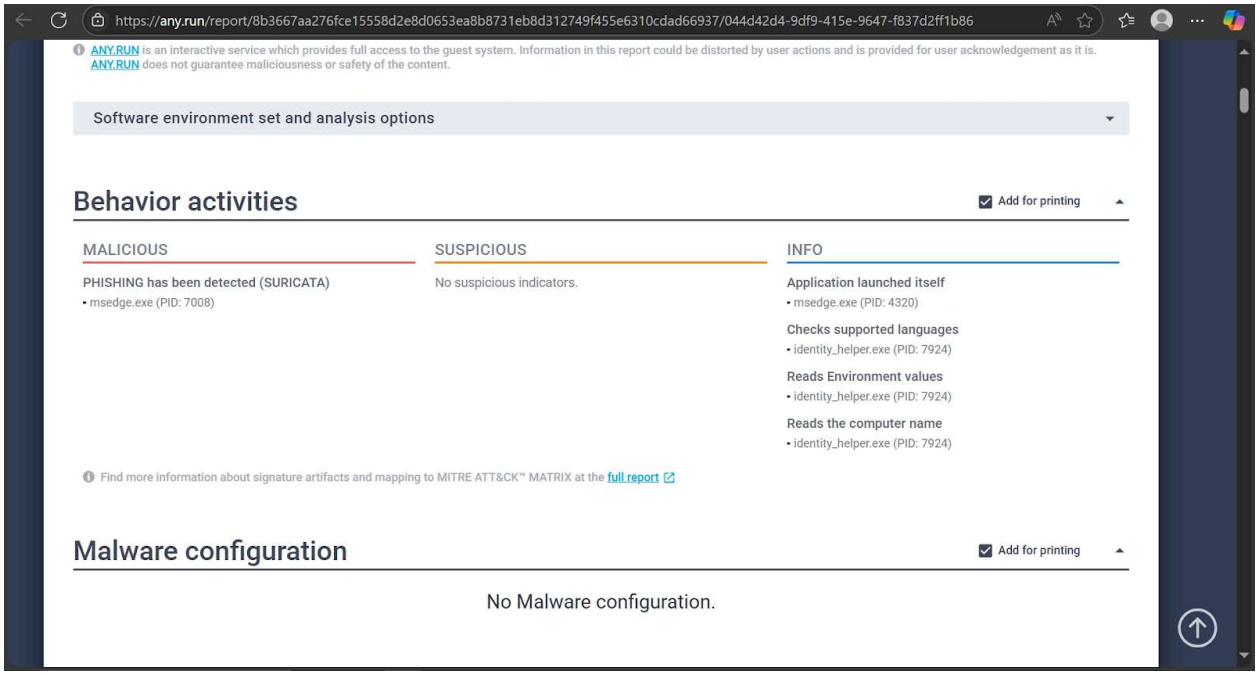


For **non-malicious samples**, the report explicitly states that **no threat was detected.** This means the analyzed file or URL does not exhibit any harmful behavior.



This process allows users to understand malware functionality in real time and is an excellent tool for analyzing both known and unknown threats in a controlled, interactive environment

ANY.RUN Malware Sample Analysis and Report Development

1. Step-by-Step Process for Analyzing a Malware Sample

1.1 Sample Preparation

- **Select Samples:** Obtain one known malicious file and one clean sample (preferably of the same type e.g., executable or document).
- **Verify File Integrity:** Generate and compare hash values (MD5, SHA1, SHA256).
- **Record Metadata:** Note the file name, size, format, and suspected behavior.

1.2 Sandbox Configuration in ANY.RUN

- **Login:** Access ANY.RUN via a registered account.
- **Environment Setup:** Choose appropriate OS (e.g., Windows 10) and analysis duration (typically 5–15 minutes).
- **Network Settings:** Enable internet simulation or restrict communication as needed.

1.3 Uploading and Executing the Sample

- **Upload Files:** Submit both the malicious and clean sample individually.
- **Execution Mode:** Choose manual interaction to simulate user behavior (e.g., opening a file, clicking dialog boxes).
- **Monitoring Tools Enabled:** Ensure process monitor, file system monitor, registry tracker, and network inspector are active.

1.4 Real-Time Behavioral Analysis

- **Process Activities:** Observe process creation, child processes, command-line arguments, and privileges.
- **Registry Modification:** Track any changes in registry keys that could indicate persistence mechanisms.
- **File System Changes:** Note newly created files, changes in system directories, and suspicious configurations.

1.5 Network Traffic Inspection

- **Traffic Monitoring:** Capture all outbound connections, DNS queries, HTTP/HTTPS requests, and C2 communication.
- **Indicators:** Look for signs like encrypted traffic, contact with suspicious domains/IPs, or data exfiltration.

2. Developing a Technical Report: Methodology Overview

Creating a technical report involves documenting findings in a structured, comparative, and actionable format.

2.1 Executive Summary

- Provide a concise summary of threat level, key malicious behaviors, and impact.

2.2 Sample Information

- **Sample Name**
- **File Hashes (MD5, SHA1, SHA256)**
- **Type, Size, Submission Date**

2.3 Behavioral Comparison (Malicious vs Clean)

Category	Malicious Sample	Clean Sample
Process Tree	Suspicious child process, privilege escalation	Normal app behavior
File Changes	Dropped executables in system folders	Temporary files only
Registry	Persistence keys added	No significant changes
Network	Contacted external C2 servers	Local communication only

2.4 Indicators of Compromise (IOCs)

- **File Paths:** e.g., `C:\Windows\System32\malware.exe`
- **Registry Keys:** e.g., `HKCU\Software\Microsoft\Windows\CurrentVersion\Run`
- **IP/Domain:** e.g., `192.168.1.105`, `malicious-site.net`

2.5 Mitigation Recommendations

- Block domains and IPs listed in the IOC section.
- Monitor file system and registry for similar changes.
- Deploy behavioral detection rules (e.g., using YARA or Sigma).
- Educate users on avoiding suspicious files and phishing tactics.

2.6 Risk Assessment

- **Threat Level:** Medium/High (based on activity and persistence).
- **Impact:** Data exfiltration, service disruption.

- **Response Priority:** Immediate mitigation and further investigation required.

3. Conclusion

The ANY.RUN analysis allows a comprehensive understanding of malware behavior. By comparing malicious and clean samples, one can distinguish between benign anomalies and real threats. A systematic approach to documenting these behaviors ensures effective communication and response planning.

References

- ANY.RUN Malware Sandbox: https://any.run
- MITRE ATT&CK Framework: https://attack.mitre.org
- VirusTotal Intelligence Platform: https://www.virustotal.com
- Malware Analysis Techniques – Practical Malware Analysis, Sikorski & Honig (2012)
- SANS Institute – Malware Analysis Reports: https://www.sans.org